

Figure 1: The 12-Primitive Crystal Lattice $3^3 \times 4^5 \times 5^4 = 17,280,000$ states
 $3^3 = 27$

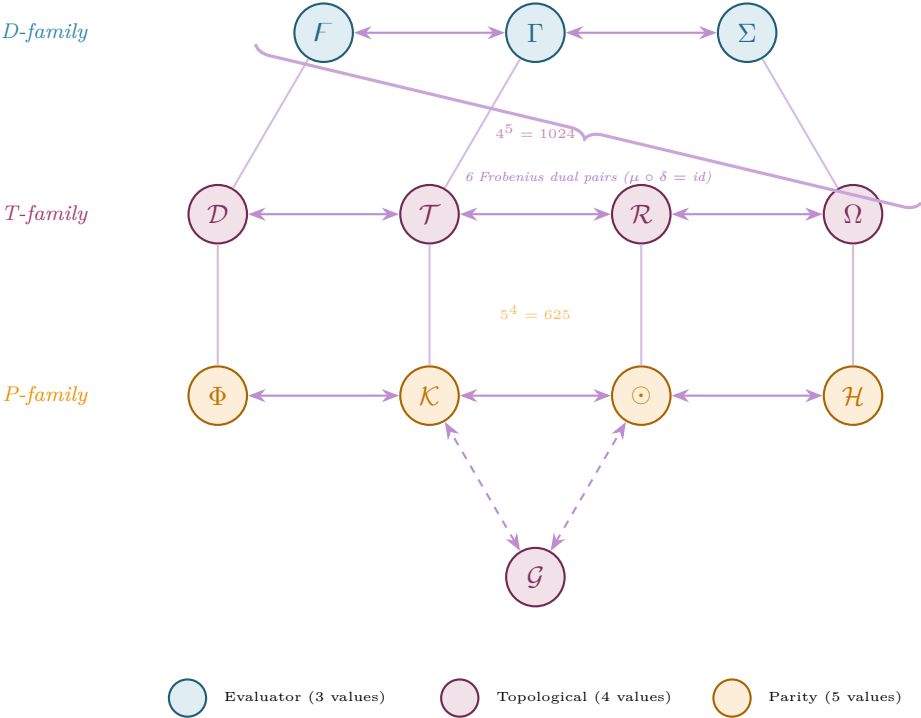


Figure 2: The Belnap FOUR Bilattice — Paraconsistent Kernel

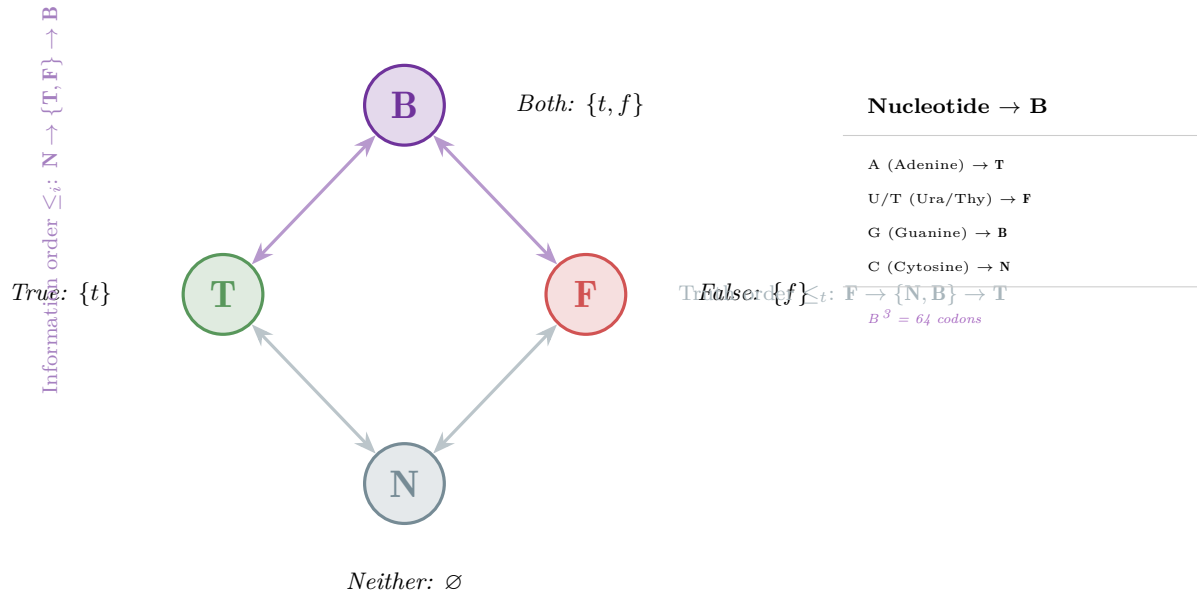


Figure 3: The 7-Stage Gene $\rightarrow$ Protein Pipeline with Primitive Transitions

B flow: $N \rightarrow T \rightarrow B \rightarrow F \rightarrow T \rightarrow T$

$\Delta(\text{DNA, Quaternary}) = 4, 0$

Stage 0: DNA GENE

$H:\odot\odot \quad \Sigma:1 \quad \Omega:\mathbb{Z}$

$D:\text{tri} \quad T:\sqcap \quad R:\leftrightarrow \quad \Phi:\sim \quad F:\ell \quad K:\text{slow} \quad \Gamma:\sqsupset \quad G:\blacktriangleright \quad \odot:<c$

transcription

$T:\sqcap \rightarrow \subset, R:\leftrightarrow \rightarrow \rightarrow$
 $K:\text{slow} \rightarrow \text{mod}, \Omega:\mathbb{Z} \rightarrow 0$

Stage 1: PRE-mRNA

$\Sigma:1 \quad \Omega:0$

$D:\text{tri} \quad T:\subset \quad R:\rightarrow \quad \Phi:\emptyset \quad F:\ell \quad K:\text{mod} \quad \Gamma:\sqsupset \quad G:\blacktriangleright \quad \odot:<c \quad H:\odot$

splicing

$R:\rightarrow \rightarrow \dagger$

Stage 2: MATURE mRNA

$H:\odot \quad \Sigma:1 \quad \Omega:0$

$D:\text{tri} \quad T:\subset \quad R:\dagger \quad \Phi:\emptyset \quad F:\ell \quad K:\text{slow} \quad \Gamma:\sqsupset \quad G:\blacktriangleright \quad \odot:<c$

translation

$R:\dagger \rightarrow \rightarrow, K:\text{slow} \rightarrow \text{fast}$
 $H:\odot \rightarrow \odot\odot, \Sigma:1 \rightarrow \neq$

Stage 3: NASCENT POLYPEPTIDE

$G:\blacktriangleright \quad \odot:<c \quad H:\odot\odot \quad \Sigma:\neq \quad \Omega:0$

$D:\text{tri} \quad T:\bowtie \quad R:\rightarrow \quad \Phi:\emptyset \quad F:\ell \quad K:\text{fast} \quad \Gamma:\sqsupset$

folding I (collapse)

$D:\text{tri} \rightarrow \odot, T:\bowtie \rightarrow \odot$
 $\Phi:\emptyset \rightarrow \sim, \Sigma:\neq \rightarrow n$

Stage 4: SECONDARY STRUCTURE

$G:\blacktriangleright \quad \odot:<c \quad H:\odot\odot \quad \Sigma:n \quad \Omega:0$

$D:\odot \quad T:\odot \quad R:\dagger \quad \Phi:\sim \quad F:\ell \quad K:\text{mod} \quad \Gamma:\sqsupset$

folding II (packing)

$R:\dagger \rightarrow \leftrightarrow, \Gamma:\sqsupset \rightarrow \mathbb{N}$
 $G:\blacktriangleright \rightarrow \wedge, \Sigma:n \rightarrow 1$

Stage 5: TERTIARY STRUCTURE

$G:\wedge \quad \odot:<c \quad H:\odot\odot \quad \Sigma:1 \quad \Omega:0$

$D:\odot \quad T:\odot \quad R:\leftrightarrow \quad \Phi:\emptyset \quad F:\ell \quad K:\text{slow} \quad \Gamma:\mathbb{N}$

assembly

$\Phi:\emptyset \rightarrow \sim, H:\odot\odot \rightarrow \odot$
 $\Sigma:1 \rightarrow \neq, \Omega:0 \rightarrow \mathbb{Z}$

Stage 6: QUATERNARY STRUCTURE

$G:\wedge \quad \odot:<c \quad H:\odot \quad \Sigma:\neq \quad \Omega:\mathbb{Z}$

$D:\odot \quad T:\odot \quad R:\leftrightarrow \quad \Phi:\sim \quad F:\ell \quad K:\text{slow} \quad \Gamma:\sqsupset$

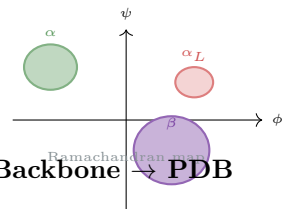


Figure 4: The Serpent Rod Folding Engine — B → Ramachandran → 3D Backbone → PDB

Input: DNA/RNA Sequence AUG UGG UGC UAC UUU AUU CAC ...

1. B EVALUATION

Nucleotide → B : A→T, U→F, G→B, C→N
 Codon triplets → B winding path ($B^3 = 64$ states)

16 transition pairs

2. B → RAMACHANDRAN MAPPING

Transition $(b_i, b_{i+1}) \rightarrow (\phi, \psi)$ per residue
 16 B transition types → 4 secondary structure classes

(ϕ, ψ) basins

3. 3D BACKBONE RECONSTRUCTION

Internal → Cartesian: place_atom algorithm
 N, C α , C, O per residue (bond lengths/angles from Pauling 1951)

N-C α -C geometry

4. CONTACT PREDICTION + ENERGY

C α -C α < 8 Å contacts $E = E_{LJ} + E_{HB} + E_{elec}$
 Primitive activation: 6-10 of 12 $\mu \circ \delta = id$ verified

Frobenius closed

Output: PDB Structure File (v3.3)

